

Composite Structures

Assignment

Due date: 13-11-2025

Write a MATLAB computer program to solve for the following problems.

Each group needs to submit one PDF document with a small write up about the methodology and your results (tables, figures etc.). See the attached excel sheet for the group formation.

Anyone from the group can send the PDF document and the MATLAB code to my email address prasunjana@gmail.com latest by 13th November, 2025.

Group	Problem Type (Levy Type Solution)	Laminate Sequence	Boundary Conditions
A	Bending of a specially orthotropic plate: 1. Calculate the deflection at the mid-point ($a/2, b/2, 0$) and stress components at ($a/2, b/2, h/2$) and ($a/4, b/4, -h/4$) of a square plate subjected to UDL. 2. Plot the mid-point deflection vs aspect ratio for $a/b=1$ to $a/b=4$. (Normalize the results as it is done in the class note).	$(0^\circ/90^\circ)_s$	SCSF
B	Free vibration of a specially orthotropic plate: 1. Calculate the first TWO frequencies of a square plate. Plot the mode shapes. 2. Plot first frequency vs aspect ratio for $a/b=1$ to $a/b=5$. (Normalize the results as it is done in the class note).	$(0^\circ/90^\circ/0^\circ)_s$	SFSF
C	Free vibration of a specially orthotropic plate: 1. Calculate the first TWO frequencies of a square plate. Plot the mode shapes. 2. Plot first frequency vs aspect ratio for $a/b=1$ to $a/b=5$. (Normalize the results as it is done in the class note).	$(90^\circ/0^\circ)_s$	SCSF

D	Bending of a specially orthotropic plate: 1. Calculate the deflection at the mid-point $(a/2, b/2, 0)$ and stress components at $(a/2, b/2, h/2)$ and $(a/4, b/4, -h/2)$ of square plate subjected to UDL. 2. Plot the mid-point deflection vs aspect ratio for $a/b=1$ to $a/b=4$. (Normalize the results as it is done in the class note).	$(90^\circ/0^\circ/90^\circ)$	SFSF
E	Free vibration of a specially orthotropic plate: 1. Calculate the first TWO frequencies of a square plate. Plot the mode shapes. 2. Plot first frequency vs aspect ratio for $a/b=1$ to $a/b=5$. (Normalize the results as it is done in the class note).	$(90^\circ/0^\circ/90^\circ)_s$	SSSF
F	Free vibration of a specially orthotropic plate: 1. Calculate the first TWO frequencies of a square plate. Plot the mode shapes. 2. Plot first frequency vs aspect ratio for $a/b=1$ to $a/b=5$. (Normalize the results as it is done in the class note).	$(0^\circ/90^\circ/0^\circ)$	SCSS

✓ Each layer is of equal thickness. Total laminate thickness is h .

✓ Material Properties: $E_{11} = 18 E_{22}$, $G_{12} = 0.8 E_{22}$, $\nu_{12} = 0.28$.